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Chapter 1

Brownian Motion in extreme lubrication

1.1 Introduction

Let us now turn to the second part of my thesis, and dare I say, the heaviest part of my very own doctoral work, both from the theoretical and the numerical standpoints.

When considering Brownian suspensions, one must remain aware of hydrodynamic interactions at play in the system. Indeed, even at long range, hydrodynamics play a strong role on the behaviour of colloids diffusing in the fluid. Thus, strong care must be taken to ensure that these interactions are accounted for at every range, including the sub-radius range, when a colloid is either close to a confining plane or another colloid. Together, we will see that typically, excellent approximations exist for the so-called far-field and moderate-field regimes, but that the near-field regime requires extra care and involves much more complex physics.

1.1.1 Colloid-plane lubrication theory

Consider now an infinite silica plane, which we will take normal to the vertical z axis, laying at $z = 0$. Over that plane is a fluid of density ρ , viscosity η , and thermal energy $k_B T$ with k_B Boltzmann's constant. As per usual Stokesian assumptions, we will take the fluid as incompressible.

In that fluid, we now submerge a colloidal silica particle of radius a , of the micron scale. The goal is now to accurately simulate such a system in the overdamped regime. Due to the no-slip boundary conditions at contact with the plane and the colloid's surface, we must modify the effective viscosity of the fluid as the colloid evolves in the z direction. This correction brings not only inhomogeneity along the z -axis but also anisotropy, as the x and y components of the friction coefficient will differ from the z component.

In his seminal work[5], Brenner derives the exact expression for the effective viscosity felt by a sphere moving perpendicularly to a plane wall, $\eta_\perp(z)$, as

$$\eta_{\perp}(z) = \eta \frac{4}{3} \sinh(\alpha) \sum_{n=1}^{\infty} \frac{n(n+1)}{(2n-1)(2n+3)} \left[\frac{2 \sinh((2n+1)\alpha) + (2n+1) \sinh(2\alpha)}{4 \sinh^2((n+\frac{1}{2})\alpha) - (2n+1)^2 \sinh^2(\alpha)} - 1 \right], \quad (1.1)$$

with $\alpha = \cosh^{-1}(\frac{z}{a})$. Because Brenner obtains the effective viscosity from full Stokes derivations, it is worth noting that this expression is exact and valid for all values of $z/a > 1$, that is to say up until colloid/plane contact. However, for the sake of efficient numerical computations later on, we will avoid infinite-series computations and prefer the use of Padé approximants [1], which were computed by Honig [8], Bevan and Prieve [3] as

TODO: Add the petit O and domain of validity.

$$\eta_{\perp}(z) \simeq \eta \frac{6z^2 + 2az}{6z^2 - 9az + 2a^2}. \quad (1.2)$$

TODO: Add Maxou-like comparison of the formulae maybe ? idk i dont fuking feel like it right now bye.

The parallel effective viscosity, on the other hand, was derived by Faxén [7] and is given by the following expression:

$$\eta_{\parallel}(z) \simeq \frac{\eta}{1 - \frac{9}{16} \frac{a}{z} + \frac{1}{8} \left(\frac{a}{z}\right)^3 - \frac{45}{256} \left(\frac{a}{z}\right)^4 - \frac{1}{16} \left(\frac{a}{z}\right)^5}. \quad (1.3)$$

These expressions suffice for the simulation of a colloid over an infinite plane in far-field to moderate-field regimes.

Forces acting upon the colloid

The Fokker-Plank equation

A much more subtle effect contributing to preserving the system's accurate physics is the so-called stochastic drift. This drift is a consequence of the effective viscosity of the fluid being inhomogeneous and anisotropic. This means that the friction coefficient γ felt by the particle will also be inhomogeneous and anisotropic, and so will be the diffusion coefficient $D = k_B T / \gamma$. In turn, we cannot consider the stochastic force $\vec{F}_{\text{stoch}}$ as a simple white noise, as it will have to be multiplicative, and thus, we will have to add a drift term to the Langevin equation in order to preserve the correct equilibrium distribution of the system.

To derive this stochastic drift contribution to the total of all forces acting upon the colloid, we start from the one-dimensional Fokker-Plank equation, which describes the evolution of the Probability Density Function (PDF) $P(X_t, X_0; t)$ of the particle's position X_t at time t , given its initial position X_0 at time $t = 0$:

$$dX_t = u(X_t)dt + g dB_t, \quad (1.4)$$

where $u(X_t)$ is the drift velocity term, g is the magnitude of the noise, and dB_t is the increment of a Wiener process.

We now take the average value of a function obeying the PDF $P(X_t, X_0; t)$, which we will call $f(X_t)$. This average value will, by definition, follow [9]:

$$\langle f(X_t) \rangle = \int f(X_t) P(X_t, X_0; t) dX_t, \quad (1.5)$$

with the initial condition $P(X_t, X_0; 0) = \delta(X_t - X_0)$, where δ is the Dirac delta function. Expanding $f(X_t)$ in a Taylor series yields:

$$f(X_t + dX_t) = f(X_t) + f'(X_t)dX_t + \frac{1}{2}f''(X_t)(dX_t)^2 + \dots \quad (1.6)$$

Substituting dX_t from the Fokker-Plank equation and taking the average, we get:

$$\left\langle \frac{df(X_t)}{dt} \right\rangle \simeq \frac{\partial f(X_t)}{\partial X_t} u(X_t) + \frac{1}{2} \frac{\partial^2 f(X_t)}{\partial X_t^2} g^2 \quad (1.7)$$

Combining this with the definition of the average value of $f(X_t)$, we can derive the Fokker-Plank equation for the PDF $P(X_t, X_0; t)$:

$$\int dX_t \frac{\partial f(X_t)}{\partial t} f(X_t) = \frac{\partial f(X_t)}{\partial X_t} u(X_t) + \frac{1}{2} \frac{\partial^2 f(X_t)}{\partial X_t^2} g^2 \quad (1.8)$$

$$= \int dX_t P(X_t, X_0; t) G f(X_t), \quad (1.9)$$

where G is the Generator, defined as $G = u(X_t) \frac{\partial}{\partial X_t} + \frac{1}{2} g^2 \frac{\partial^2}{\partial X_t^2}$. Taking the adjoint of G as $G^\dagger$, we can rewrite the above equation as:

$$\int dX_t P(X_t, X_0; t) G f(X_t) = \int dX_t f(X_t) G^\dagger P(X_t, X_0; t). \quad (1.10)$$

We can then derive the Fokker-Plank equation for the PDF $P(X_t, X_0; t)$ as:

$$\frac{\partial P(X_t, X_0; t)}{\partial t} = G^\dagger P(X_t, X_0; t) = -\frac{\partial}{\partial X_t} (u(X_t) P(X_t, X_0; t)) + \frac{1}{2} \frac{\partial^2}{\partial X_t^2} (g^2 P(X_t, X_0; t)). \quad (1.11)$$

Thus, for a bulk Brownian motion with Langevin equation

$$\frac{dx}{dt} = \sqrt{2D_0} \eta(t), \quad (1.12)$$

with x the colloid's position, D_0 the bulk diffusion coefficient, and $\eta(t)$ a white noise of zero mean and unit variance, we can identify $u(x) = 0$ and $g = \sqrt{2D_0}$, and thus, the Fokker-Plank equation for the PDF of the colloid's position reads:

$$\frac{\partial P(x, x_0; t)}{\partial t} = D_0 \frac{\partial^2 P(x, x_0; t)}{\partial x^2}, \quad (1.13)$$

which is the well-known diffusion equation.

Fokker-Plank equation with multiplicative noise

Now, in the case of an inhomogeneous and anisotropic diffusion coefficient, the Fokker-Plank equation reads

$$dX_t = u(X_t)dt + g(X_t)dB_t , \quad (1.14)$$

$$X_{t+\Delta t} = X_t + \int_t^{t+\Delta t} u(X_s)ds + \int_t^{t+\Delta t} g(X_s)dB_s , \quad (1.15)$$

$$\int_t^{t+\Delta t} g(X_s)dB_s ds \equiv B([1 - \alpha]X_t + \alpha X_{t+\Delta t}) \Delta W(t) , \quad (1.16)$$

$$\Delta W(t) = \int_t^{t+\Delta t} dB_s , \quad (1.17)$$

$$X_{t+\Delta t} = X_t + g(X_t)\Delta W(t) \quad (1.18)$$

$$\int_t^{t+\Delta t} g(X_s)dB_s ds = g(X_t + \alpha[X_{t+\Delta t} - X_t]) \Delta W(t) \quad (1.19)$$

$$\approx g(X_t + \alpha g(X_t)\Delta W(t)) \Delta W(t) \quad (1.20)$$

$$\approx g(X_t)\Delta W(t) + \alpha g(X_t)g'(X_t)[\Delta W(t)]^2 . \quad (1.21)$$

$$X_{t+\Delta t} = X_t + u(X_t)\Delta t + g(X_t)\Delta W(t) + \alpha g(X_t)g'(X_t)[\Delta W(t)]^2 , \quad (1.22)$$

$$\frac{\partial P(X_t, t)}{\partial t} = -\frac{\partial}{\partial X_t} K_1(X_t)P(X_t, t) + \frac{1}{2} \frac{\partial^2}{\partial X_t^2} K_2(X_t)P(X_t, t) , \quad (1.23)$$

$$K_1(X_t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta X_t}{\Delta t} \quad (1.24)$$

$$= u(X_t) + \alpha g(X_t)g'(X_t) , \quad (1.25)$$

and

$$K_2(X_t) = \lim_{\Delta t \rightarrow 0} \frac{(\Delta X_t)^2}{\Delta t} \quad (1.26)$$

$$= g(X_t)^2 . \quad (1.27)$$

why the fuck z ?

$$\frac{\partial P(X_t, t)}{\partial t} = -\frac{\partial}{\partial X_t} \{u(X_t) + \alpha g(X_t)g'(X_t)\} P(X_t, t) + \frac{1}{2} \frac{\partial^2}{\partial X_t^2} \{g(X_t)^2\} P(X_t, t) . \quad (1.28)$$

Derivation of the stochastic drift term

$$dz = \frac{F(z)}{\gamma_{\perp}(z)} dt + \sqrt{2D_{\perp}(z)} dB_t, \quad (1.29)$$

where $F(z)$ is the total of deterministic forces and $F(z) = -k_B T \ln(U(z))$.

$$\left\langle \frac{\Delta z}{\Delta t} \right\rangle \equiv \bar{v}_d = \frac{F(z)}{\gamma_{\perp}(z)} + \alpha \frac{dD_{\perp}(z)}{dz}. \quad (1.30)$$

$$F_{\text{eq}} = - \frac{dU_{\text{steady-state}}(z)}{dz} \quad (1.31)$$

$$U_{\text{steady-state}}(z) = -k_B T \ln(P_{\text{steady-state}}(z)), \quad (1.32)$$

$$\frac{d}{dz} \left\{ -u(z) - \alpha g(z)g'(z) + \frac{1}{2} \frac{d}{dz} g(z)^2 \right\} P_{\text{steady-state}}(z) = 0. \quad (1.33)$$

$$(-u(z) - \alpha g(z)g'(z))P_{\text{steady-state}}(z) + \frac{1}{2} \frac{\partial}{\partial z} \{g(z)^2 P_{\text{steady-state}}(z)\} = 0, \quad (1.34)$$

We may now substitute $Y = g^2(z)P_{\text{steady-state}}(z)$, which yields

$$-\frac{u(z) + \alpha g(z)g'(z)}{g^2(z)/2} Y + \frac{dY}{dz} = 0, \quad (1.35)$$

which solves as

$$P_{\text{steady-state}}(z) = \frac{1}{g^2(z)} \exp \left\{ 2 \int_0^z \frac{u(z') + \alpha g(z')g'(z')}{g^2(z')} dz' \right\}. \quad (1.36)$$

$$U_{\text{steady-state}}(z) = -k_B T \int^{\Delta t} \frac{u(z') + (\alpha - 1)g(z')g'(z')}{g^2(z')/2} dz'. \quad (1.37)$$

$$\frac{F_{\text{eq}}}{\gamma_{\perp}(z)} = \frac{F(z)}{\gamma_{\perp}(z)} + (\alpha - 1)g(z)g'(z) \quad (1.38)$$

$$\bar{v}_d = \frac{1}{\gamma_{\perp}(z)} \frac{dU_{\text{steady-state}}(z)}{dz} + \frac{dD_{\perp}(z)}{dz}. \quad (1.39)$$

$$v_d = \frac{k_B T}{\gamma_{\perp}(z)} \left[-\frac{1}{l_D} B \exp \left(-\frac{z}{L_D} \right) + \frac{1}{l_B} \right]. \quad (1.40)$$

$$v_{\text{stochastic drift}} = 2D_0 a \frac{2a^2 + 12az + 21z^2}{(2a^2 + 9az + z^2)^2}. \quad (1.41)$$

TODO: Absolutely verify every single equation maxime wrote. they might be wrong.

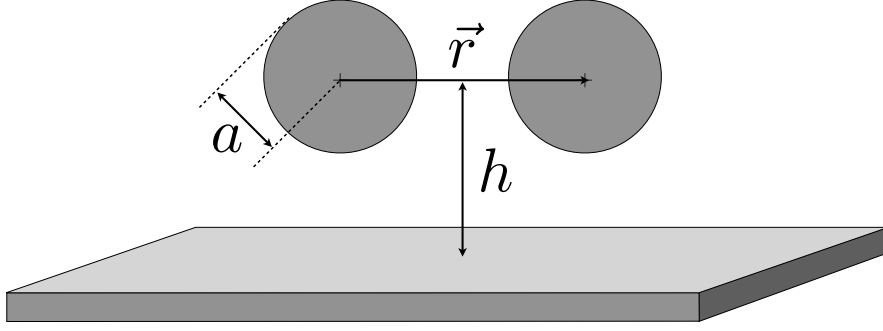


Figure 1.1: Experimental setup of Dufresne et al. [6] for the study of two colloids diffusing over a plane wall.

Simple Langevin simulation of a confined colloid

We may now write the Euler-discretised equations of motion for a colloid confined in the z direction as:

$$x_i = x_{i-1} + \sqrt{2D_{\parallel}(z_{i-1})\Delta t}w_{x_i}, \quad (1.42)$$

$$z_i = z_{i-1} + \bar{v}_d(z_{i-1})\Delta t + \sqrt{2D_{\perp}(z_{i-1})\Delta t}w_{z_i}, \quad (1.43)$$

where w_{x_i} and w_{z_i} are two independent Gaussian white noises of zero mean and Δt variance, and where $\bar{v}_d(z)$ is the average drift velocity of the colloid at position z as computed in Eq. 1.41, which includes both deterministic and stochastic contributions as per equation 1.30.

1.1.2 Colloid-colloid lubrication theory

[2]

1.1.3 Coupling colloid-wall and colloid-colloid lubrication theories

Naturally, the next step is to couple both lubrication effects together. Luckily, such study has already been carried out both theoretically and experimentally by Dufresne et al. [6]. In this article, the authors discuss the modification on colloid cross-diffusion due to wall and colloid-colloid-induced lubrication effects. The following setup, depicted in Figure 1.1.3, is used: two colloids of radius $a = 0.485 \pm 0.025 \mu\text{m}$ (Duke Scientific lot 21024) are submerged in a water-salt solution of 2mM [NaCl] and viscosity $\eta = 0.817\text{cP}$. The water-salt layer is $140 \pm 2 \mu\text{m}$ thick, and sealed between a microscope slide and a No. 1 coverslip, with the system fully sealed with UV-cured epoxy to prevent evaporation of the fluid and fluid flows in the bulk. The system is maintained at $T = 29.00 \pm 0.01^\circ\text{C}$. In these conditions, the colloid self-diffusion is found to be $D_0 = \frac{k_B T}{6\pi\eta a} = 0.550 \pm 0.028 \mu\text{m}^2/\text{s}$.

The authors then trap the two colloids with a single optical tweezer, by alternating the position of the focal plane at the two positions of interest at a frequency of 200Hz

using a galvanometer-driven mirror. Then, by blocking the beam for $t = 83\text{ms}$, they retrieve the sphere positions over five video frames only to ensure minimal out-of-focal displacement. From there on, authors carry out experiments over a grid of starting positions in (r, h) space, with r the center-to-center distance between colloids and $2\mu\text{m} < r < 10\mu\text{m}$, and h the colloid center-to-wall distance, with $1\mu\text{m} < h < 30\mu\text{m}$. By reactivating the trapping, they can repeat the experiment with perfect reproducibility for each starting position of interest in their system.

As discussed in the previous sections, colloids in a same bath should see their motion coupled through hydrodynamic interactions, and the coupling to the wall itself should also modify their expected diffusion. To investigate these coupling effects, the authors decompose the trajectories of the colloids into cooperative and relative motions, respectively $\vec{\rho} = \vec{r}_1 + \vec{r}_2$ and $\vec{r} = \vec{r}_1 - \vec{r}_2$ along ($\parallel$) and perpendicular ($\perp$) directions to the line of centers of the colloids. From there, they fill displacement histograms for each mode of motion at each of the five time steps taken into consideration and retrieve the effective diffusion coefficients from the standard deviation of each histogram, as

$$\langle \Delta r_{i,\alpha}(\tau) \Delta r_{j,\beta} \rangle = 2D_{i\alpha,j\beta}\tau, \quad (1.44)$$

with $i, j \in \{1, 2\}$ and $\alpha, \beta \in \{\parallel, \perp\}$, where τ is the elapsed time, and where $\langle \cdot \rangle$ denotes an ensemble average.

A typical result of the experiment for a given set of (r, h) is shown in Figure 1.1.3. There, MSDs are extracted from displacement histograms at each of the five time lags into consideration. From there, authors can extract the effective cross-diffusion coefficient.

For comparisons purposes with their experimentally retrieved cross-diffusions in the four possible modes, the authors then compute by different means the theoretical expected value of the cross-diffusion coefficients. In the case of two colloids in the bulk that are separated by a sufficient distance to approximate them by point forces, the flow field around one of the colloids can be described by the fundamental solution to Stokes' equations for a point force, the Stokeslet [10], and the Green's function for the flow at $\vec{x}$ in the α direction from a unit force at $\vec{r}_j$ in the β direction reads :

$$G_{\alpha\beta}^S(\vec{x} - \vec{r}_j) = \frac{1}{8\pi\eta} \left(\frac{\delta_{\alpha\beta}}{|\vec{x} - \vec{r}_j|} + \frac{(\vec{x} - \vec{r}_j)_\alpha (\vec{x} - \vec{r}_j)_\beta}{|\vec{x} - \vec{r}_j|^3} \right). \quad (1.45)$$

Then, for the case of two colloids of same radius, they recover the following expression for the cross-diffusion coefficient in the bulk:

$$\frac{D_{\perp}^{C,R}}{2D_0} = 1 \pm \frac{3a}{4r} + \mathcal{O}\left(\frac{a^3}{r^3}\right), \quad (1.46)$$

$$\frac{D_{\parallel}^{C,R}}{2D_0} = 1 \pm \frac{3a}{2r} + \mathcal{O}\left(\frac{a^3}{r^3}\right), \quad (1.47)$$

where positive and negative corrections correspond respectively to the collective and relative modes of motion. These expressions are known as the Oseen approximations

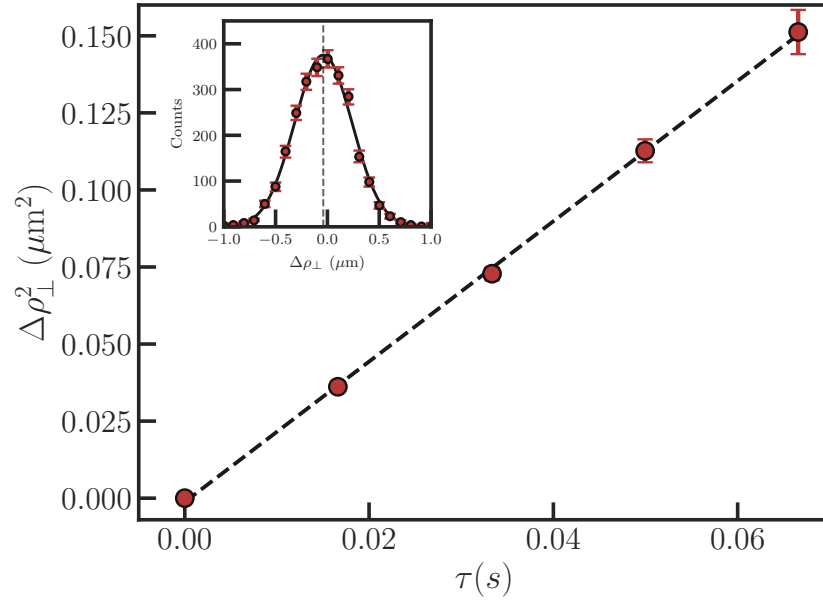


Figure 1.2: Adapted from [6]. Measured pair diffusion for the relative perpendicular mode of motion of the two colloids. The upper inset shows the histogram of all binned displacements for a given time step, for $h = 25.5 \pm 0.7 \mu\text{m}$ and $r = 7.00 \pm 0.25 \mu\text{m}$. This process is repeated over all time steps and each histogram is fitted with a Gaussian curve to recover the root mean square displacement $\Delta\rho(r, h, \tau)$ to recover the MSD of the main plot. The effective cross-diffusion coefficient for the perpendicular mode is then extracted from the slope of the MSD.

[10], and are valid solely in the far-field regime due to the point-force approximation not taking into account the finite-size effects.

Now, in the presence of a wall, one must take extra care to ensure no-slip boundary conditions at the wall as well as no fluid penetration through the wall itself. To do this, one may use the single-colloid method of images [4], where a Stokeslet (S), as Source Doublet (D) and a Stokeslet Doublet (SD) of opposite strength to the Stokeslet of interest at $\vec{r}_j$ above the wall is placed at equal distance below the wall, at $\vec{R}_j = \vec{r}_j - 2h\vec{e}_z$. The goal of the mirror Stokeslet is to cancel most of the normal penetration at the wall. The Source Doublet ensures no-slip at the wall, and the Stokeslet Doublet cancels remaining penetration. Then, the flow at the position of the image colloid is written

$$G_{\alpha\beta}^W(\vec{x} - \vec{R}_j) = -G_{\alpha\beta}^S(\vec{x} - \vec{R}_j) + 2h^2 G_{\alpha\beta}^D(\vec{x} - \vec{R}_j) - 2h G_{\alpha\beta}^{SD}(\vec{x} - \vec{R}_j) , \quad (1.48)$$

with

$$G_{\alpha\beta}^D(\vec{x} - \vec{R}_j) = \frac{1}{8\pi\eta} (1 - 2\delta_{\beta z}) \frac{\partial}{\partial R_{j,\beta}} \left(\frac{(\vec{x} - \vec{R}_j)_\alpha}{|\vec{x} - \vec{R}_j|^3} \right) , \quad (1.49)$$

$$G_{\alpha\beta}^{SD}(\vec{x} - \vec{R}_j) = (1 - 2\delta_{\beta z}) \frac{\partial}{\partial R_{j,\beta}} G_{\alpha z}^S(\vec{x} - \vec{R}_j) . \quad (1.50)$$

which yields the two following modes of motion for the cross-diffusion coefficients

$$\frac{D_z}{D_0} = 1 - \frac{9}{8} \frac{a}{r} + \mathcal{O}\left(\frac{a^3}{r^3}\right) , \quad (1.51)$$

$$\frac{D_{xy}}{D_0} = 1 - \frac{9}{16} \frac{a}{r} + \mathcal{O}\left(\frac{a^3}{r^3}\right) . \quad (1.52)$$

One may notice the resemblance between the correction terms in the above expressions and the corrections in the Faxén formula from equation 1.3, which is not surprising as both corrections are due to the same wall-induced hydrodynamic effects. However, these expressions are only valid in the far-field regime, and fail to capture the lubrication effects in the near-field regime, which is precisely the regime of interest for our simulations.

The authors then propose an intermediate regime, first by naively superimposing the Oseen and Blake approximations as $D_\psi^{-1}(r, h) = D_\psi^{-1}(r) + [D_{xy}^{-1}(h) - D_0^{-1}] / 2$ (in dash lines in Figure 1.1.3), which is a crude approximation which fails to capture the behaviour of the cross-diffusion for colloids below 50 radii from the plane. The authors then propose that the colloid must interact with its image, its neighbour, and its neighbour's image. In that case, the cross-diffusion is given by

$$\frac{D_{\perp}^{C,R}}{2D_0} = 1 - \frac{9}{16} \frac{a}{r} \pm \frac{3}{4} \frac{a}{r} \left(1 - \frac{1 + \frac{3}{2}\xi}{(1 + \xi)^{3/2}} \right) \quad (1.53)$$

$$\frac{D_{\parallel}^{C,R}}{2D_0} = 1 - \frac{9}{16} \frac{a}{r} \pm \frac{3}{2} \frac{a}{r} \left(1 - \frac{1 + \xi + \frac{3}{2}\xi^2}{(1 + \xi)^{5/2}} \right) \quad (1.54)$$

with $\xi = 4h^2/r^2$ and up to $\mathcal{O}\left(\frac{a^3}{r^3}\right)$ and $\mathcal{O}\left(\frac{a^3}{h^3}\right)$ (in solid lines in Figure 1.1.3).

Furthermore, as noted by the authors, in the very far-field regime, the Oseen approximations of Eq. 1.47 should be recovered numerically and experimentally, whereas they expect the above expressions to hold in the very near-field regime as the wall hinders coupling between colloid motions.

In the following part, we will go about confirming the above predictions numerically, but we will push the study much deeper into the lubrication regimes to investigate its validity once we cannot ignore finite-size effects anymore in the lubrication approximations.

1.2 The lubrication-corrected RigidBlobWalls algorithm for simulating Brownian suspensions in confinement

1.2.1 Lubrication-corrected mobility tensor

At this level, it becomes more convenient to work in terms of *self-mobility* and *self-resistance* matrices rather than in terms of effective viscosities. The mobility matrix $\mathbf{M}$ relates the velocity $\vec{U}$ of the particle to the applied force $\vec{F}$ via $\vec{U} = \mathbf{M} \cdot \vec{F}$, while the resistance matrix $\mathbf{R}$ is its inverse, relating the force to the velocity via $\vec{F} = \mathbf{R} \cdot \vec{U}$. In the context of a sphere near a plane wall, these matrices become anisotropic and depend on the distance z from the wall.

1.2.2 Numerical methods for simulating dense Brownian suspensions in close confinement

Naturally, researchers have been pondering on how to simulate dense Brownian suspensions for a few decades already [2]. However, the numerical complexity of such a method (and namely, how to efficiently solve for a multiplicative noise and, a fortiori, how to compute efficiently the square root of a diffusion or mobility tensor without computational cost blowups).

1.2.3 Application to the Dufresne experiments: going beyond the Blake approximation

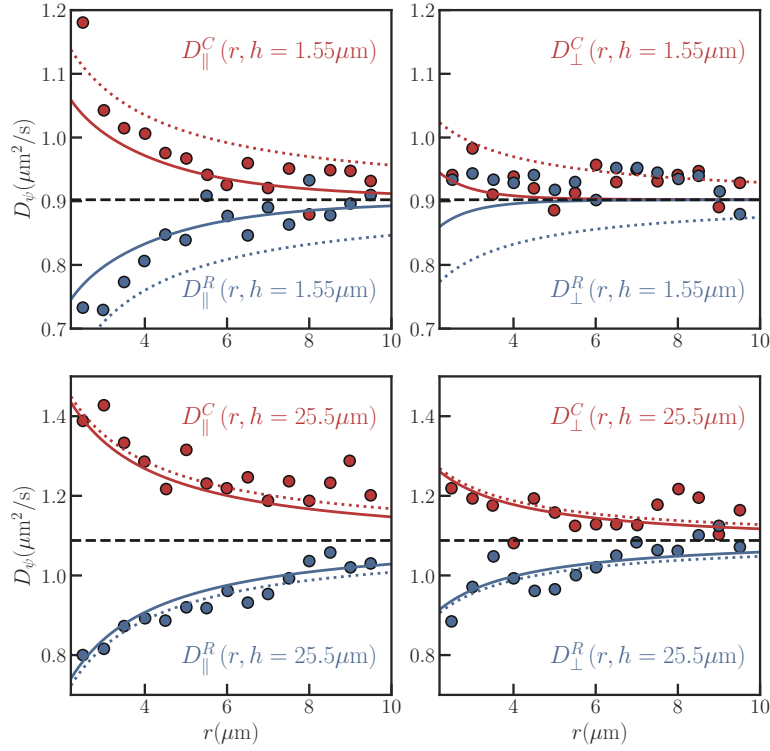


Figure 1.3: Adapted from [6]. Measured cross-diffusion coefficients for the relative perpendicular and parallel modes of motion of the two colloids, as a function of the center-to-center distance r between the colloids, for different values of the center-to-wall distance h . The dash lines correspond to the naive superposition of Oseen and Blake approximations, while the solid lines correspond to the more accurate approximation taking into account the interaction with the image colloid.

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